

Imaris Analysis Comparison of Cluster vs. Non-Cluster Plastic Fluorescence Metrics

Imaris Fluorescence Metric Comparison

The following is a comprehensive breakdown of fluorescence intensity metrics, including quality, mean intensity, max intensity, and voxels. The intent is to compare at two levels - cluster and non-cluster plastics, and across different images as well.

The Goal

The intent is to be able to visualize these metrics so that a (somewhat) universal filter can be established to help quantify plastic uptake in the CHT, while also excluding out clustered plastics that may result from macrophage uptake.

All source code for the following analysis scripts can be found [here](#)

```
voxel_compares <- list(
  c("Image 1", "Image 14"),
  c("Image 1", "Image 28"),
  c("Image 14", "Image 28")
)
final_table %>%
  ggplot(aes(x = Image_ID, y = Voxel, fill = Image_ID,
color = Image_ID)) +
  geom_boxplot(outlier.shape = NA, alpha = 0.6) +
  geom_jitter(width = 0.2, alpha = 0.4, size = 1) +
  facet_wrap(~Spot_Category) +
  labs(
    title = "Plastics Distribution across Images",
    x = "Voxel",
    y = "freq"
  ) +
  theme_prism(base_size = 14) +
  scale_fill_prism(palette = "prism_light") +
  scale_color_prism(palette = "prism_light") +
  stat_compare_means(
    comparisons = voxel_compares,
    method = "t.test",
```

```
label = "p.signif"
)
```

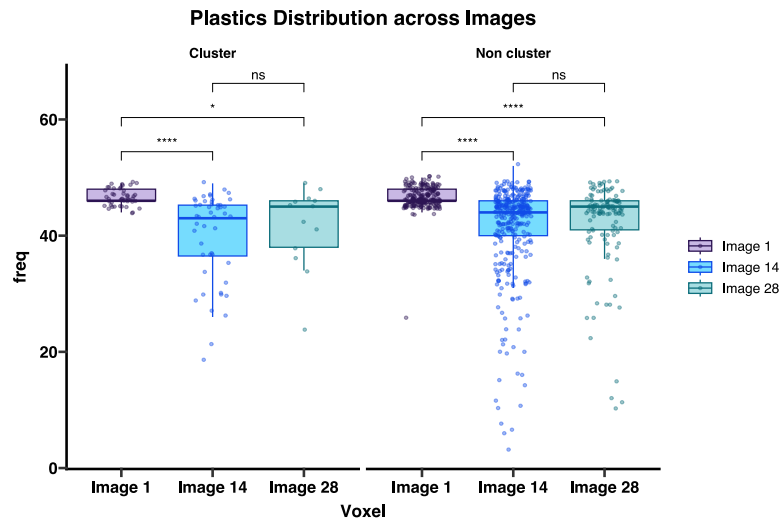


Figure 1: Voxels per spot for clustered plastics across images

If we zoom in to Image 1 - which is a *kdrl:mCherry - stab2 crispant* zebrafish injected with **50nm plastics**, the voxel comparison looks like the following:

```
voxel_image1_compare <- list(c("Cluster", "Non
cluster"))

final_table %>%
  filter(Image_ID == "Image 1") %>%
  ggplot(aes(
    x = Spot_Category,
    y = Voxel,
    color = Spot_Category,
    fill = Spot_Category
  )) +
  geom_boxplot(outlier.shape = NA, alpha = 0.6) +
  geom_jitter(width = 0.2, alpha = 0.4, size = 1) +
  labs(
    title = "Cluster vs. NonCluster Voxel in Image 1",
    x = "Spot Category",
    y = "Voxel"
  ) +
  theme_prism(base_size = 14) +
  scale_fill_prism(palette = "prism_light") +
```

```

scale_color_prism(palette = "prism_light") +
stat_compare_means(
  comparisons = voxel_image1_compare,
  method = "t.test",
  label = "p.signif"
)

```

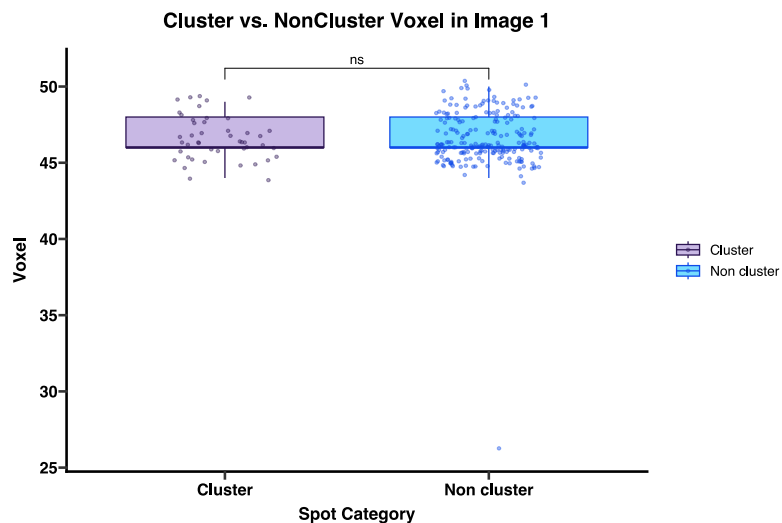


Figure 2: Voxels per spot for clustered vs. nonclustered plastics in Image 1

If we put these comparisons together side by side, we then observe the voxel differences between the two:

```

voxel_compares <- list(
  c("Image 1", "Image 14"),
  c("Image 1", "Image 28"),
  c("Image 14", "Image 28")
)
final_table %>%
  ggplot(aes(x = Image_ID, y = Voxel, fill = Image_ID, color = Image_ID)) +
  geom_boxplot(outlier.shape = NA, alpha = 0.6) +
  geom_jitter(width = 0.2, alpha = 0.4, size = 1) +
  facet_wrap(~Spot_Category) +
  labs(
    title = "Plastics Distribution across Images",
    x = "Voxel",
    y = "freq"
  ) +
  theme_prism(base_size = 14) +

```

```

scale_fill_prism(palette = "prism_light") +
scale_color_prism(palette = "prism_light") +
stat_compare_means(
  comparisons = voxel_compares,
  method = "t.test",
  label = "p.signif"
)

voxel_image1_compare <- list(c("Cluster", "Non cluster"))

final_table %>%
  filter(Image_ID == "Image 1") %>%
  ggplot(aes(
    x = Spot_Category,
    y = Voxel,
    color = Spot_Category,
    fill = Spot_Category
  )) +
  geom_boxplot(outlier.shape = NA, alpha = 0.6) +
  geom_jitter(width = 0.2, alpha = 0.4, size = 1) +
  labs(
    title = "Cluster vs. NonCluster Voxel in Image 1",
    x = "Spot Category",
    y = "Voxel"
  ) +
  theme_prism(base_size = 14) +
  scale_fill_prism(palette = "prism_light") +
  scale_color_prism(palette = "prism_light") +
  stat_compare_means(
    comparisons = voxel_image1_compare,
    method = "t.test",
    label = "p.signif"
  )

```

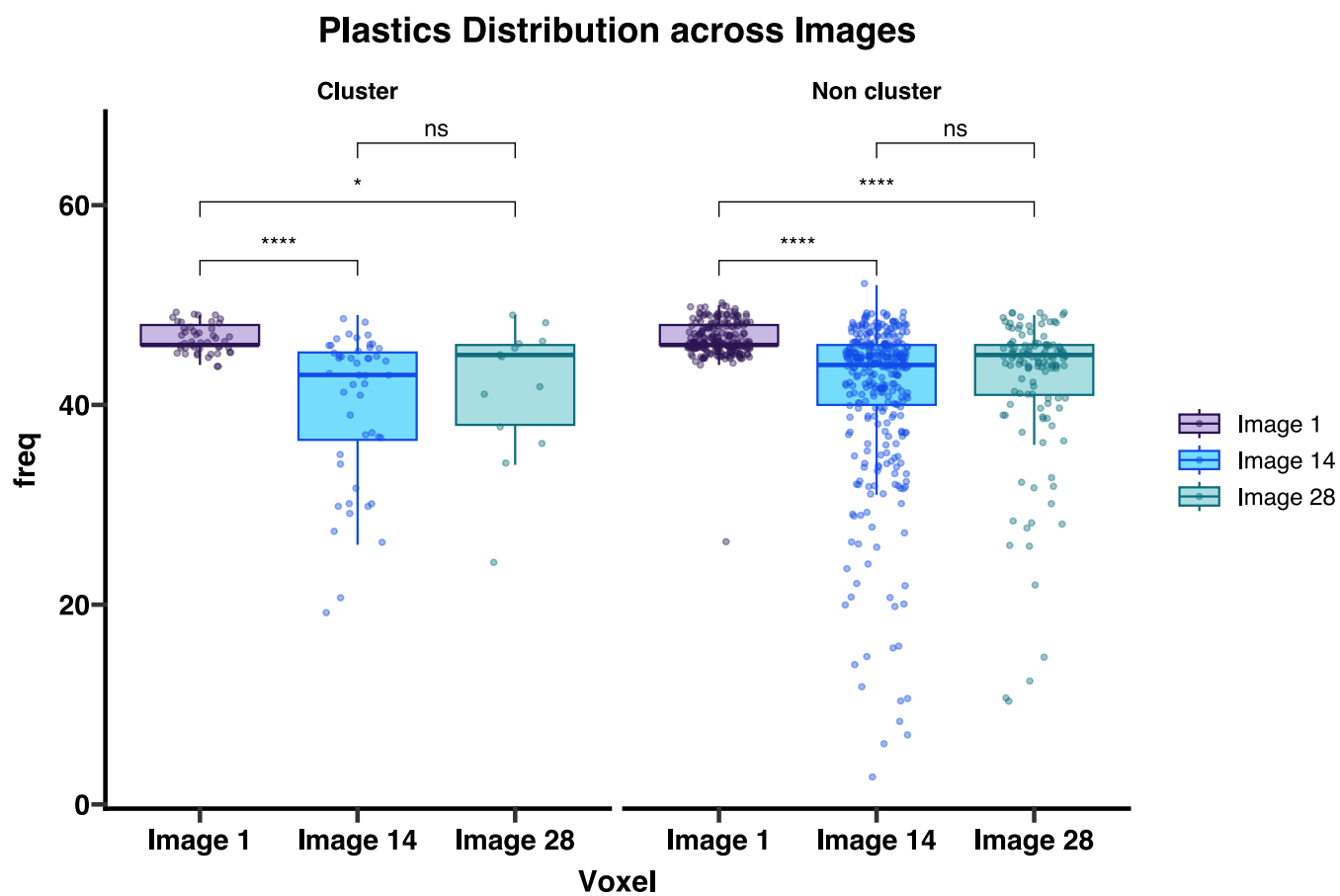


Figure 3: Voxels per spot for clustered plastics across all images

Cluster vs. NonCluster Voxel in Image 1

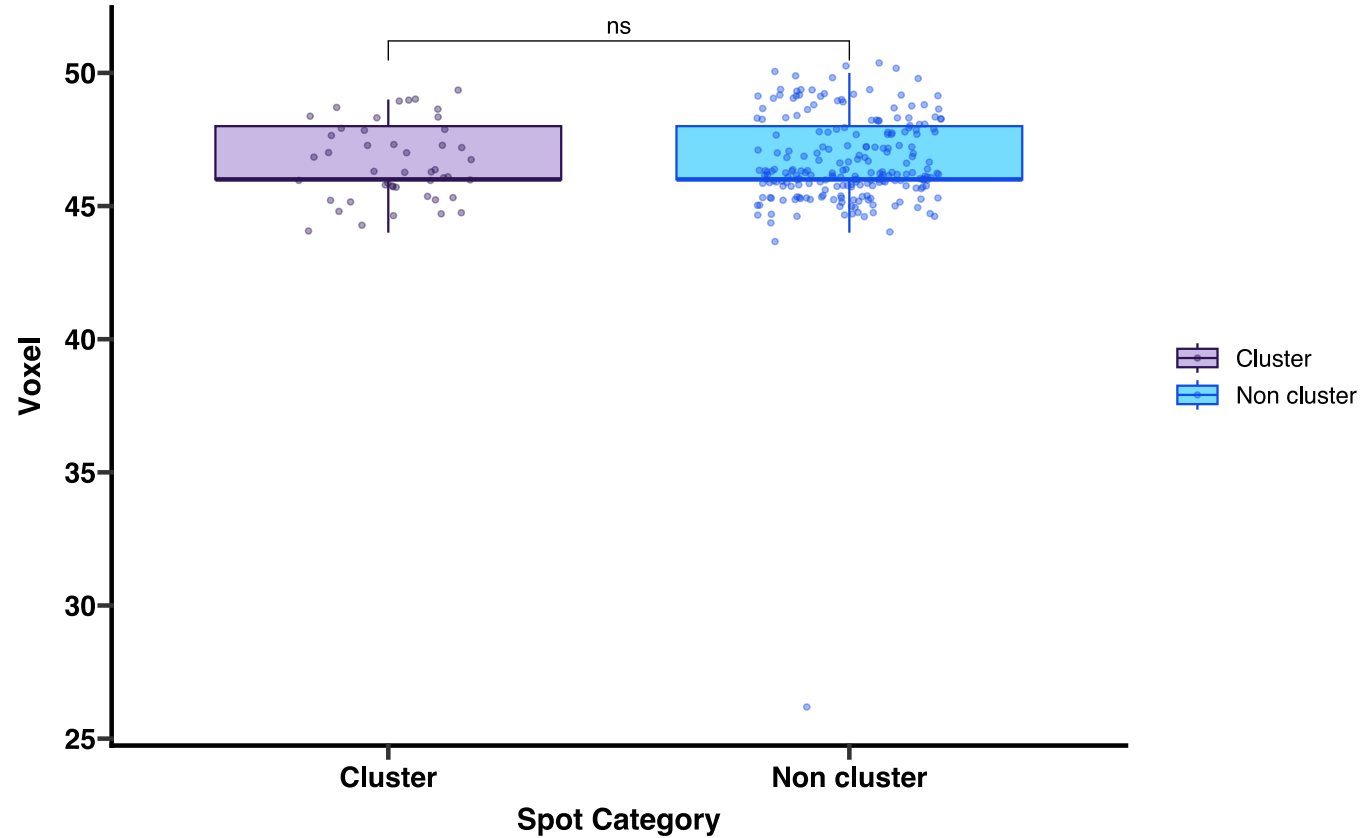


Figure 4: Voxels per spot for clustered vs. non-clustered within Image 1